Points: 18.25/20 (91.25%)
well done! ☺

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Serie 1

Aufg 1.

(i) t.s. $\mathbb{Z}[\sqrt{-1}] \leq \mathbb{C}$ is a subring.

proof. Since $\sqrt{-1}$ is the square-root of $-1 \in \mathbb{Z}$, it holds that

$$\mathbb{Z}[\sqrt{-1}] = \{a + b\sqrt{-1} \mid a, b \in \mathbb{Z}\} \quad (1)$$ why? easy but not obvious

Thus $1 = 1 + 0 \in \mathbb{Z}[\sqrt{-1}]$. Then let $x = a_1 + b_1\sqrt{-1}$, $y = a_2 + b_2\sqrt{-1} \in \mathbb{Z}[\sqrt{-1}]$.

Then it holds that

$$x - y = \underbrace{(a_1 - a_2)}_{\in \mathbb{Z}} + \underbrace{(b_1 - b_2)}_{\in \mathbb{Z}}\sqrt{-1} \in \mathbb{Z}[\sqrt{-1}]$$

$$x \cdot y = \underbrace{(a_1a_2 - b_1b_2)}_{\in \mathbb{Z}} + \underbrace{(a_1b_2 + a_2b_1)}_{\in \mathbb{Z}}\sqrt{-1} \in \mathbb{Z}[\sqrt{-1}]$$

which implies that $\mathbb{Z}[\sqrt{-1}]$ is a subring. ∎ 1.75/2

(ii) t.s. $\mathbb{Z}[\sqrt{-1}]$ is a free abelian group & $\mathbb{Z}[\sqrt{-1}] = \mathbb{Z} + \mathbb{Z}\sqrt{-1}$ ✓

proof By (i) it follows that $\mathbb{Z}[\sqrt{-1}]$ is an abelian group. (since a ring)

claim $(1, \sqrt{-1})$ is a basis of $\mathbb{Z}[\sqrt{-1}]$.

Since (1) one sees that $(1, \sqrt{-1})$ generates $\mathbb{Z}[\sqrt{-1}]$. why?

Then let $a_1, a_2 \in \mathbb{Z}$ s.t. $a_1 + a_2\sqrt{-1} = 0$. It follows that $a_2\sqrt{-1} = -a_1$ which implies $a_1, a_2 = 0$. (otherwise $\sqrt{-1} \in \mathbb{Q}$ which is a contradiction) Thus the independency follows and therefore the claim.

In particular, we get $\mathbb{Z}[\sqrt{-1}] = \mathbb{Z} + \mathbb{Z}\sqrt{-1}$. why? ∎

From now on, we can identify $\mathbb{Z}[\sqrt{-1}]$ as $\mathbb{Z}^2$. how? 1.5/2

(iii) t.s. if $u \in \mathbb{Z}[\sqrt{-1}]^\times \Rightarrow |u| = 1$.

proof Let $u \in \mathbb{Z}[\sqrt{-1}]^\times$ be an unit.

claim. $|\cdot|$ is multiplicative.

So there exists a $v \in \mathbb{Z}[\sqrt{-1}]^\times$ s.t. $u \cdot v = 1$ and by claim we get

$$1 = |u \cdot v| = \underbrace{|u| \cdot |v|}_{\in \mathbb{Z}^+_{>0} \in \mathbb{Z}_{>0}}$$

$|u| = \sqrt{\mathrm{Re}(u)^2 + \mathrm{Im}(u)^2}$ is not an integer in general, ex $|1+i| = \sqrt{2} \notin \mathbb{Z}_{>0}$

Thus $|u| = 1$. It is left to prove the claim.

Let $x = (a_1, b_1) \in \mathbb{Z}^2 \cong \mathbb{Z}[\sqrt{-1}]$, $y = (a_2, b_2) \in \mathbb{Z}[\sqrt{-1}]$. Then

$$|x \cdot y| = \sqrt{(a_1a_2 - b_1b_2)^2 + (a_1b_2 + a_2b_1)^2}$$

$$= a_1^2a_2^2 - 2a_1a_2b_1b_2 + b_1^2b_2^2 + a_1^2b_2^2 + 2a_1b_2a_2b_1 + a_2^2b_1^2$$

$$= \sqrt{(a_1^2 + b_1^2)} \cdot \sqrt{(a_2^2 + b_2^2)} = |u| \cdot |v|$$

and we can conclude. ∎ 1.5/2

iv) t.s. $\mathbb{Z}[\sqrt{-1}]^\times = \{\pm 1, \pm\sqrt{-1}\}$

proof Let $u = a + b\sqrt{-1} \in \mathbb{Z}[\sqrt{-1}]^\times$. Then by (iii) we get

$1 = |u| = \sqrt{a^2 + b^2}$, and since $a^2, b^2 \in \mathbb{Z}^+_{>0}$ we get

$(a^2 = 1, b^2 = 0)$ or $(a^2 = 0, b = 1)$ thus $(a = 1, b = 0)$ or $(a = 0, b = 1)$

which implies $u \in \{\pm 1, \sqrt{-1}\}$ and the inclusion $\mathbb{Z}[\sqrt{-1}]^\times \subseteq \{\pm 1, \pm\sqrt{-1}\}$ follows.

Since $(-1)^2 = 1$, $(\pm\sqrt{-1})^2 = 1$ we get the equality. ∎

In particular, this proof shows that $\mathbb{Z}[\sqrt{-1}]^\times = \{u \in \mathbb{Z}[\sqrt{-1}] \mid N(u) = 1\}$

2/2

Aufg 1.)

v) We denote $v = (2+\sqrt{-1})(2-\sqrt{-1})$.

t.s. $v = 5$

proof We denote $x = 2+\sqrt{-1} \in \mathbb{Z}[\sqrt{-1}]$.

Then one see's that $v = x \cdot \bar{x} = N(x) = 2^2+1 = 5$. ☺ 1/1

vi) Let $a,b \in \mathbb{Z}[\sqrt{-1}]$ s.t. $3 = ab$.

t.s. $a \in \mathbb{Z}[\sqrt{-1}]^\times$ or $b \in \mathbb{Z}[\sqrt{-1}]^\times$.

proof Since $3 = ab$, it holds that $9 = N(ab) = N(a) \cdot N(b)$.

$N(x) = Re(x)^2 + Im(x)^2$

Therefore $N(a) = 3$ could also occour. How do we exclude it?

Since $N(a), N(b) \in \mathbb{Z}_{\geq 0}$ and 3 is irreducible in $\mathbb{Z}$, we get $N(a) \in \mathbb{Z}^\times$ or $N(b) \in \mathbb{Z}^\times$, thus $N(a) = 1$ or $N(b) = 1$.

By the proof of 6a(iv) this implies $a \in \mathbb{Z}[\sqrt{-1}]^\times$ or $b \in \mathbb{Z}[\sqrt{-1}]^\times$ and we can conclude.

1.5/2

vii) t.s. $\mathbb{Z}[\sqrt{-1}] \cap \mathbb{Q} = \mathbb{Z}$. ✓

proof Since $\mathbb{Z} \subseteq \mathbb{Z}[\sqrt{-1}]$, $\mathbb{Z} \subseteq \mathbb{Q}$ Ring ext. we get $\mathbb{Z} \subseteq \mathbb{Z}[\sqrt{-1}] \cap \mathbb{Q}$.

Let $x \in \mathbb{Z}[\sqrt{-1}] \cap \mathbb{Q}$. Then there exist $a,b \in \mathbb{Z}$ s.t.

$$x = a + b\sqrt{-1} \in \mathbb{Q}$$

Furthermore, there exist $m,n \in \mathbb{Q}$, $n \neq 0$ s.t.

$$a + b\sqrt{-1} = \frac{m}{n} \Rightarrow b\sqrt{-1} = \frac{m-an}{n}$$ ✓

If $b \neq 0$ we get $\sqrt{-1} = \frac{m-an}{bn} \in \mathbb{Q}$ which is absurd, hence $b = 0$ and therefore $x = a$. This implies $x \in \mathbb{Z}$ hence, $\mathbb{Z}[\sqrt{-1}] \cap \mathbb{Q} \subseteq \mathbb{Z}$ and the equality follows.

well done!

2/2

Aufg 2.)

Let R be a ring.

t.s. R is noetherian $\Leftrightarrow$ Any ideal of R is finitely generated.

proof "$\Rightarrow$"

We assume there exists an ideal $I \subseteq R$ which is not finitely generated. Then let $a_1 \in I$. By assumption, we get $I \neq (a_1)$. Thus we can find $a_2 \in I \setminus (a_1)$ and we have $(a_1) \subsetneq (a_1, a_2) \subsetneq I$. By continuing this inductively we get a infinite chain

$$(a_1) \subsetneq (a_1, a_2) \subsetneq _____ \subsetneq I \subseteq R$$

which implies that R is not noetherian.

"$\Leftarrow$"

Let $(\mathfrak{a}_i)_i$ be a non-decreasing chain of ideals of R.

Then we define the ideal

$$I = \bigcup_{i=1}^{\infty} \mathfrak{a}_i \subseteq R, \quad \text{(why is it an ideal?)}$$

and by assumption there exist $a_1, \dots, a_n \in I$ s.t. $I = (a_1, \dots, a_n)$.

Then for each $i \in \{1, \dots, n\}$ there exists a $k_i \in \mathbb{N}$ s.t. $a_i \in \mathfrak{a}_{k_i}$

where we may assume (up to reordering) that $m = k_n = \max_{i \in \{1,\dots,n\}} (k_i)$,

thus $(a_1, \dots, a_n) \subseteq \mathfrak{a}_m$.

Then for any $j \geq n$ it holds that

$$I = (a_1, \dots, a_n) \subseteq \mathfrak{a}_m \subseteq \mathfrak{a}_j \subseteq I$$

which implies $\mathfrak{a}_m = \mathfrak{a}_j$ and therefore is this chain static ✓

and we can conclude. ☑ 4/4

Aufgabe 3

(i) Let $I \neq 0$, $I \subseteq \mathbb{Z}[\sqrt{-1}]$ be an ideal.

t.s. I is a free $\mathbb{Z}$-Module of rank 2.

proof. Since $\mathbb{Z}^2 \cong \mathbb{Z}[\sqrt{-1}]$ (iso ?) we know that $\mathbb{Z}[\sqrt{-1}]$ is a free $\mathbb{Z}$-Module of rank 2. Since $I \subset \mathbb{Z}[\sqrt{-1}]$, I can not contain any element of finite order, hence I is a free $\mathbb{Z}$-Module as well. ✓

Let $0 \neq v_1 = a + b\sqrt{-1} \in I$.

If $a \neq 0$, we choose $v_1 = a - b\sqrt{-1}$. Otherwise we choose $v_2 = b + b\sqrt{-1} \in I$ (since $b = -b\sqrt{-1}\sqrt{-1} \in I$)

Then (v_1, v_2) are linearly independent and it holds that

$$v_1\mathbb{Z} + v_2\mathbb{Z} \subseteq I$$

Therefore we get $2 \leq \operatorname{rank}(I)$. But since $I \subseteq \mathbb{Z}^2$ we get $\operatorname{rank}(I) = 2$. 2 ✓

(ii) t.s. $\mathbb{Z}[\sqrt{-1}]$ is noetherian.

proof Let $I \subseteq \mathbb{Z}[\sqrt{-1}]$ be an ideal.

claim I is finitely generated.

we may assume $I \neq (0)$ since this case is evident.

So by (ii) there exist $v_1, v_2 \in I$ s.t. $I = v_1\mathbb{Z} + v_2\mathbb{Z}$

Since $\mathbb{Z} \subseteq \mathbb{Z}[\sqrt{-1}]$ it holds by definition that

$$I = v_1\mathbb{Z} + v_2\mathbb{Z} \subseteq (v_1, v_2) \subseteq I$$

($v_1\mathbb{Z} + v_2\mathbb{Z}$: same thing; (v_1, v_2): generated as ideal.)

Thus $I = (v_1, v_2)$ is finitely generated and by Aufgabe 2 provides us that $\mathbb{Z}[\sqrt{-1}]$ is noetherian. ✓ 1/1